

ANSWER KEY — Integration Lesson 39 Test: Parametric & Polar

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. $x = 2t$, $y = t - 3$, t in $[0, 4]$. (a) Points at $t = 0, 2, 4$. (b) Eliminate the parameter; name the curve.

Work: (a) $t = 0 \rightarrow (0, -3)$; $t = 2 \rightarrow (4, -1)$; $t = 4 \rightarrow (8, 1)$. (b) From $x = 2t$, $t = x/2$; substitute: $y = x/2 - 3$. All three points satisfy it (e.g. $8/2 - 3 = 1$ ✓). **Answer: (a) (0, -3), (4, -1), (8, 1) (b) $y = x/2 - 3$ — a straight line of slope $1/2$, traced from (0, -3) to (8, 1).**

2. $x = t^2$, $y = 8t$. (a) dx/dt and dy/dt . (b) Speed at $t = 3$.

Work: (a) $dx/dt = 2t$, $dy/dt = 8$. (b) At $t = 3$: $dx/dt = 2(3) = 6$, $dy/dt = 8$. Speed = $\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. (A 3-4-5 triangle doubled: 6-8-10.) **Answer: (a) $dx/dt = 2t$, $dy/dt = 8$ (b) speed = 10. Grader note:** the components must be evaluated at $t = 3$ BEFORE applying Pythagoras; $\sqrt{(2t)^2 + 8^2}$ left in terms of t earns partial credit only.

3. $x = 5t + 2$, $y = 12t - 1$. (a) Show the speed is constant and find it. (b) Distance travelled from $t = 1$ to $t = 4$.

Work: (a) $dx/dt = 5$ and $dy/dt = 12$ — both constants, no t appears, so the speed cannot change: speed = $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ at every time. (The +2 and -1 only shift the starting point; they vanish when differentiated.)

(b) $L = \int_1^4 13 \, dt = 13 \cdot (4 - 1) = 39$. Check with straight-line geometry: at $t = 1$ the point is (7, 11); at $t = 4$ it is (22, 47); $\Delta x = 15$, $\Delta y = 36$, distance = $\sqrt{225 + 1296} = \sqrt{1521} = 39$ ✓. **Answer: (a) speed = 13 always (b) distance = 39. Grader note:** in (b) the elapsed time is $4 - 1 = 3$, not 4 (the classic "wrong width" slip).

4. $x = 3\cos t$, $y = 3\sin t$, t in $[0, 2\pi]$. (a) Speed. (b) Distance. (c) Circumference check.

Work: (a) $dx/dt = -3\sin t$, $dy/dt = 3\cos t$. Speed = $\sqrt{9\sin^2 t + 9\cos^2 t} = \sqrt{9(\sin^2 t + \cos^2 t)} = \sqrt{9 \cdot 1} = 3$, constant (using the Pythagorean identity $\sin^2 t + \cos^2 t = 1$).

(b) $L = \int_0^{2\pi} 3 \, dt = 3 \cdot 2\pi = 6\pi$.

(c) The path is the circle of radius 3 ($x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$), whose circumference is $2\pi r = 2\pi \cdot 3 = 6\pi$ ✓ — one lap exactly. **Answer: (a) speed = 3 (b) 6π (c) matches the circumference $2\pi(3) = 6\pi$.**

5. $x = 4\cos 2t$, $y = 4\sin 2t$, t in $[0, 2\pi]$. (a) Speed. (b) Distance. (c) Why is it more than 8π ? (d) When is the first lap finished?

Work: (a) chain rule: $dx/dt = -8\sin 2t$, $dy/dt = 8\cos 2t$. Speed $= \sqrt{(64\sin^2 2t + 64\cos^2 2t)} = \sqrt{64 \cdot 1} = 8$, constant.

(b) $L = \int_0^{2\pi} 8 \, dt = 8 \cdot 2\pi = 16\pi$.

(c) The point stays on the radius-4 circle ($x^2 + y^2 = 16\cos^2 2t + 16\sin^2 2t = 16$), circumference 8π . Since $16\pi = 2 \times 8\pi$, the point goes around TWICE: the inner "2t" makes the angle sweep at double rate, and $\int \text{speed} \, dt$ counts every bit of motion, including re-traced ground.

(d) A lap is finished when the angle $2t$ reaches $2\pi \rightarrow t = \pi$. Check: from $t = 0$ to π the distance is 8π = one circumference ✓. **Answer: (a) 8 (b) 16π (c) it laps the circle twice ($16\pi = 2 \cdot 8\pi$) (d) at $t = \pi$. Grader note:** forgetting the chain-rule factor 2 gives speed 4 and distance 8π — method credit only.

6. $x = 3t^2$, $y = 2t^3$, t in $[0, \sqrt{3}]$. Exact distance travelled.

Work: $dx/dt = 6t$, $dy/dt = 6t^2$. Speed² $= 36t^2 + 36t^4 = 36t^2(1 + t^2)$, so speed $= \sqrt{(36t^2(1+t^2))} = 6t\sqrt{1+t^2}$ (valid since $t \geq 0$, so $\sqrt{(36t^2)} = 6t$).

$L = \int_0^{\sqrt{3}} 6t\sqrt{1+t^2} \, dt$. Let $u = 1 + t^2$, $du = 2t \, dt \rightarrow 6t \, dt = 3 \, du$. New limits: $t = 0 \rightarrow u = 1$; $t = \sqrt{3} \rightarrow u = 1 + 3 = 4$.

$L = 3 \int_1^4 \sqrt{u} \, du = 3 \cdot \left[\frac{2}{3} u^{3/2} \right]_1^4 = 2 \left[u^{3/2} \right]_1^4 = 2(4^{3/2} - 1^{3/2}) = 2(8 - 1) = 14$.

Checks: $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$ ✓; $d/dt[2(1+t^2)^{3/2}] = 2 \cdot (3/2)(1+t^2)^{1/2} \cdot 2t = 6t\sqrt{1+t^2}$ ✓ (the original speed). **Answer: distance = 14. Grader note:** limits must be converted to u (or the antiderivative converted back to t); using u -limits 0 and $\sqrt{3}$ gives a wrong answer.

7. (a) $r = 4$. (b) $\theta = \pi/6$. (c) (x, y) of the polar point $(4, \pi/6)$. (d) Table of $r = 2 + 2\cos \theta$; name the curve.

Work: (a) All points at distance 4 from the origin, any angle: a circle of radius 4 centred at the origin.

(b) All points at angle $\pi/6$ (30°) from due east, any distance: a ray from the origin heading up-and-right.

(c) $x = r \cos \theta = 4\cos(\pi/6) = 4 \cdot (\sqrt{3}/2) = 2\sqrt{3}$; $y = r \sin \theta = 4\sin(\pi/6) = 4 \cdot (1/2) = 2$. Check: $x^2 + y^2 = 12 + 4 = 16 = 4^2$ ✓ (distance 4, as it must be).

(d) $\theta = 0 \rightarrow r = 2 + 2(1) = 4$; $\theta = \pi/2 \rightarrow r = 2 + 2(0) = 2$; $\theta = \pi \rightarrow r = 2 + 2(-1) = 0$; $\theta = 3\pi/2 \rightarrow r = 2 + 2(0) = 2$. Fat side ($r = 4$) due east, pinching to the origin at due west: a cardioid (heart curve) pointing east. **Answer: (a) circle of radius 4 (b) ray at angle $\pi/6$ (c) $(2\sqrt{3}, 2)$ (d) $r = 4, 2, 0, 2$ — a cardioid opening east.**

8. (a) Area inside $r = 6$ for θ from 0 to 2π , checked against πr^2 . (b) Area swept for θ from 0 to $\pi/2$.

Work: (a) $A = \frac{1}{2} \int_0^{2\pi} 6^2 d\theta = \frac{1}{2} \int_0^{2\pi} 36 d\theta = \frac{1}{2} \cdot 36 \cdot 2\pi = 36\pi$. Geometry: $\pi r^2 = \pi \cdot 6^2 = 36\pi$ ✓ — the pizza-wedge machine agrees with elementary geometry.

(b) $A = \frac{1}{2} \int_0^{\pi/2} 36 d\theta = \frac{1}{2} \cdot 36 \cdot (\pi/2) = 9\pi$. Expected fraction: sweeping $\pi/2$ out of 2π is one quarter of the circle, and $36\pi/4 = 9\pi$ ✓. **Answer: (a) 36π (b) 9π , exactly one quarter of 36π .**

Grader note: the limits in polar area are ANGLES, and they control what fraction of the shape is measured.

9. Exact area inside $r = 2\cos \theta$, sweeping $-\pi/2$ to $\pi/2$; justify the limits and show every half-angle step.

Work — why these limits: $r = 2\cos \theta$ equals 0 at $\theta = -\pi/2$ and again at $\theta = \pi/2$, and it is positive in between; the whole circle is drawn exactly once over $[-\pi/2, \pi/2]$. Continuing on to 2π would re-trace the same circle and double-count every wedge.

Set up: $A = \frac{1}{2} \int_{-\pi/2}^{\pi/2} (2\cos \theta)^2 d\theta = \frac{1}{2} \int_{-\pi/2}^{\pi/2} 4\cos^2 \theta d\theta = 2 \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta$.

Half-angle identity: $\cos^2 \theta = (1 + \cos 2\theta)/2$. Verify: at $\theta = 0$, left = 1 and right = $(1 + \cos 0)/2 = (1+1)/2 = 1$ ✓; at $\theta = \pi/2$, left = 0 and right = $(1 + \cos \pi)/2 = (1 - 1)/2 = 0$ ✓.

Substitute: $2 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta)/2 d\theta = \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta$.

Antiderivative: $\theta + (\sin 2\theta)/2$ (linear-inside rule: $\int \cos 2\theta d\theta = (\sin 2\theta)/2$). Check: $d/d\theta[\theta + (\sin 2\theta)/2] = 1 + \cos 2\theta$ ✓.

Evaluate: at $\theta = \pi/2$: $\pi/2 + (\sin \pi)/2 = \pi/2 + 0 = \pi/2$. At $\theta = -\pi/2$: $-\pi/2 + (\sin(-\pi))/2 = -\pi/2 + 0 = -\pi/2$. Difference = $\pi/2 - (-\pi/2) = \pi$.

Geometry check: $r = 2\cos \theta$ is the circle of radius 1 centred at $(1, 0)$, so its area is $\pi \cdot 1^2 = \pi$ ✓.

Answer: $A = \pi$. Grader note: sweeping 0 to 2π gives 2π — exactly double, the fingerprint of double-counting.

10. (a) Student writes $A = \int_0^{2\pi} 5 \, d\theta = 10\pi$ for $r = 5$. (b) Student gets $9\pi/2$ for $r = 3\cos \theta$ instead of $9\pi/4$.

Work: (a) **Error:** the student integrated $r \, d\theta$ instead of $\frac{1}{2}r^2 \, d\theta$ — the polar slice is a pizza WEDGE, not a rectangle. A wedge of angle $d\theta$ is the fraction $d\theta/(2\pi)$ of the whole circle, so its area is $(d\theta/2\pi) \cdot \pi r^2 = \frac{1}{2}r^2 \, d\theta$. The quantity $r \, d\theta$ is the LENGTH of the little arc, not an area (note 10π even has units of length, not area).

Correct: $A = \frac{1}{2} \int_0^{2\pi} 5^2 \, d\theta = \frac{1}{2} \int_0^{2\pi} 25 \, d\theta = \frac{1}{2} \cdot 25 \cdot 2\pi = 25\pi$, matching $\pi \cdot 5^2 = 25\pi$ ✓.

(b) **Error:** here the formula is right but the LIMITS are wrong. $r = 3\cos \theta$ traces its entire circle as θ runs from $-\pi/2$ to $\pi/2$; over $[0, 2\pi]$ the curve is traced twice, so every wedge is counted twice — which is why the student's answer is exactly double the truth.

Correct: $A = \frac{1}{2} \int_{-\pi/2}^{\pi/2} (3\cos \theta)^2 \, d\theta = (9/2) \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta$. Half-angle: $\cos^2 \theta = (1 + \cos 2\theta)/2$, so this is $(9/2) \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta)/2 \, d\theta = (9/4) [\theta + (\sin 2\theta)/2]_{-\pi/2}^{\pi/2} = (9/4) [(\pi/2 + 0) - (-\pi/2 + 0)] = (9/4) \cdot \pi = 9\pi/4$.

Geometry check: $r = 3\cos \theta$ is the circle of radius $3/2$ centred at $(3/2, 0)$, area $\pi(3/2)^2 = 9\pi/4$ ✓.

(And the student's $9\pi/2$ is exactly $2 \times 9\pi/4$ ✓.) **Answer: (a) used $\int r \, d\theta$ instead of $\frac{1}{2} \int r^2 \, d\theta$ (that integrand is arc length); correct area 25π . (b) limits $[0, 2\pi]$ trace the circle twice; correct area $9\pi/4$ using θ from $-\pi/2$ to $\pi/2$.**

Grader note: full credit needs the ERROR NAMED in each part, not just the corrected number. These are the two classic polar-area mistakes: missing $\frac{1}{2}r^2$, and limits that re-trace the curve.